Lab# 05: Extract Indicators of Compromise from Security Reports

Objective: Learn to extract indicators of compromise (IOCs) from security reports.

Indicators of compromise (IOC) in cybersecurity refers to clues or evidence that suggest a network or system has been breached or attacked. For example, IOCs can be unusual network traffic behavior, unexpected software installations, user sign-ins from abnormal locations, and large numbers of requests for the same file.

IOCs encompass diverse types of data, including:

- IP addresses
- Domain names
- URLs and Email address
- Unusual Outbound Network Traffic
- Network traffic patterns
- Filenames, paths, and hash files
- Anomaly in Privileged Useraccount Activity
- Geographical Irregularities
- Log-In Red Flags
- Increases in Database Read Volume
- HTML Response Sizes
- Large Requests for the Same File
- Mismatched Port-Application Traffic
- Suspicious Registry or System File Changes
- Unusual DNS Requests
- Unexpected Patching of Systems
- Mobile Device Profile Changes
- Bundles of Data in the Wrong Place
- Web Traffic with Unhuman Behavior
- Signs of DDoS Activity

Challenges in managing IOCs effectively:

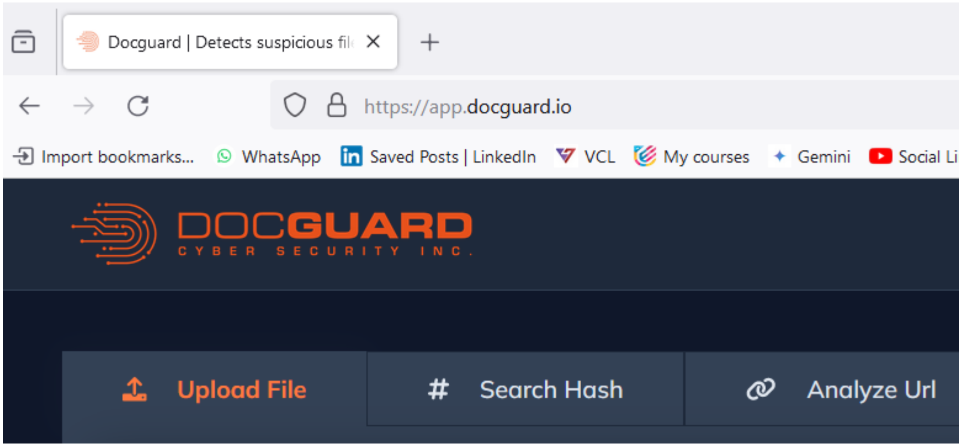
- Monitoring and analysing IOCs presents several challenges. The sheer volume alone of IOCs detected daily can overwhelm security teams, in addition to keeping IOCs up to date in today's rapidly evolving threat landscape. However, even the most effective IOC management is not enough to reduce risk of threats and prevent attacks.
- Managing IOCs is a reactive approach that relies on historical data of known threats. New, advanced threats may evade indicator-based threat detection. IOCs are more effective when combined with proactive measures to detect threats faster, such as endpoint security, real-time threat intelligence, threat management platforms, and identity access controls.

1. DOCGUARD

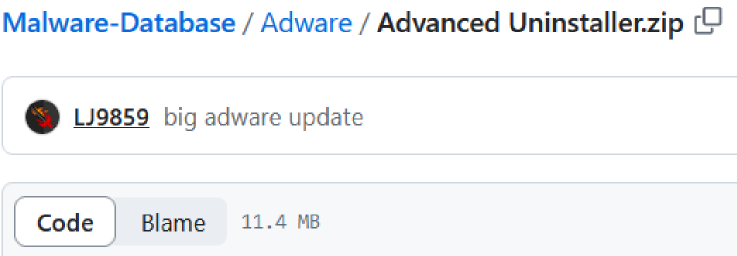
Docguard.io is a service designed to analyze malware samples, Hashes and URLS. It uses a technique called "structural analysis" to identify malicious code and functionalities within files.

- Quickly identifying malware: Docguard.io boasts fast analysis times, allowing CIT professionals to efficiently determine if a file is malicious.
- Advanced threat detection: The service claims to detect various advanced threats, including macro-based malware and obfuscated code.
- Identifying IOCs: Docguard.io can extract IOCs from analysed files, which can be helpful for further investigation and threat hunting.

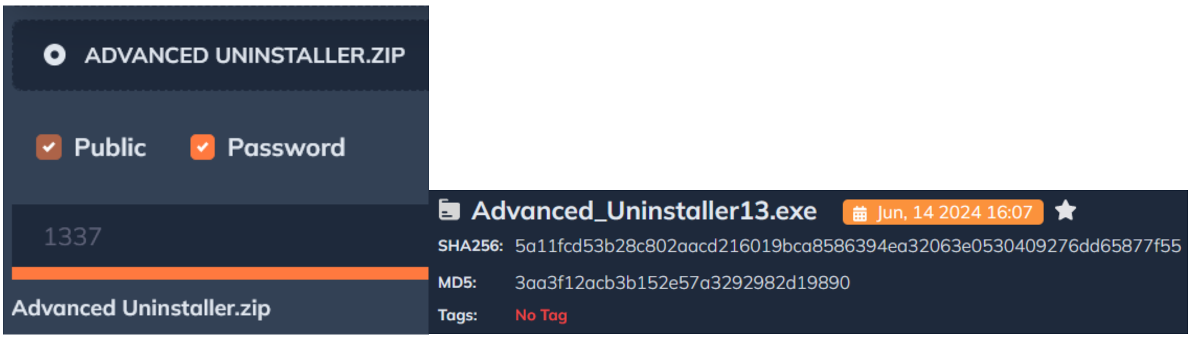
Step 1: Open https://app.docguard.io in a sandbox environment OR on a Virtual Machine.



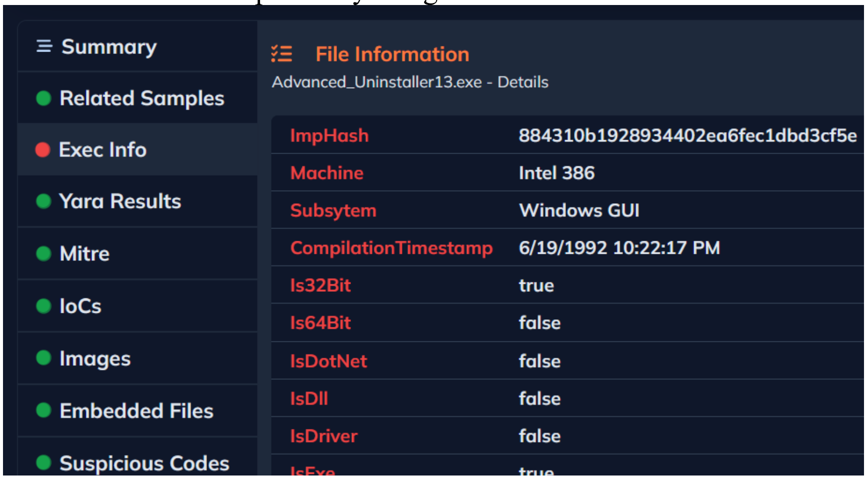
Step 2: Download malware files from https://github.com/LJ9859/Malware-Database. The password for all zip files is "1337" without the quotation marks.



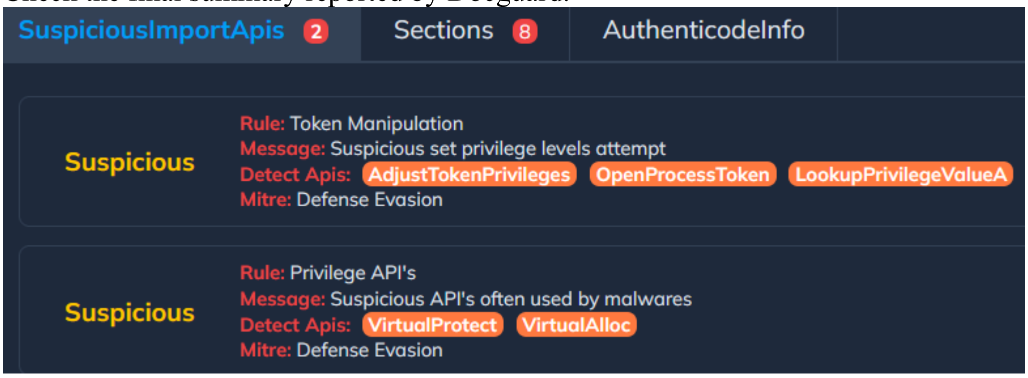
Step 3: Check Adware if it is malicious or not by uploading this to Docguard.



Step 4: Study the file information reported by Docguard.



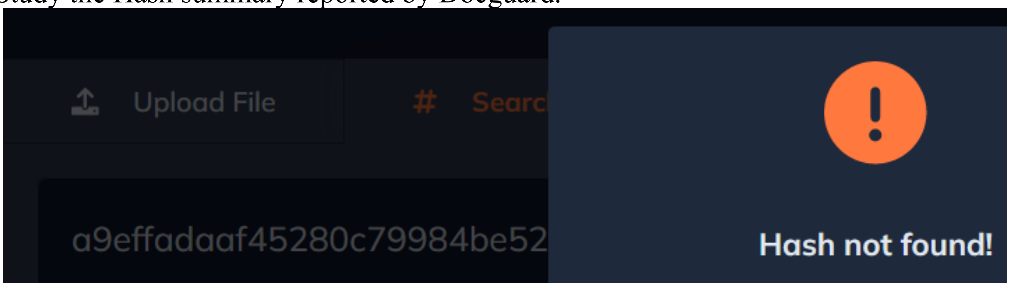
Step 5: Check the final summary reported by Docguard.



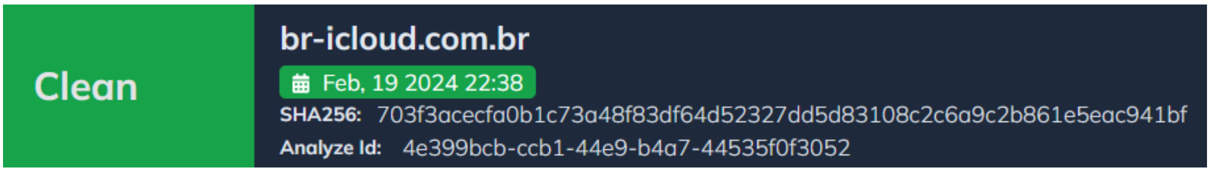
Step 6: Copy malicious file/malware Hashes from https://github.com/bitdefender/malware-ioc/blob/master/metamorfo_malware/samples.hash

Step 7: Paste into Docguard and analyze

Step 8: Study the Hash summary reported by Docguard.



Step 9: Copy URL from https://www.kaggle.com/datasets/sid321axn/malicious-urls-dataset, paste the URL into Docguard and analyze.

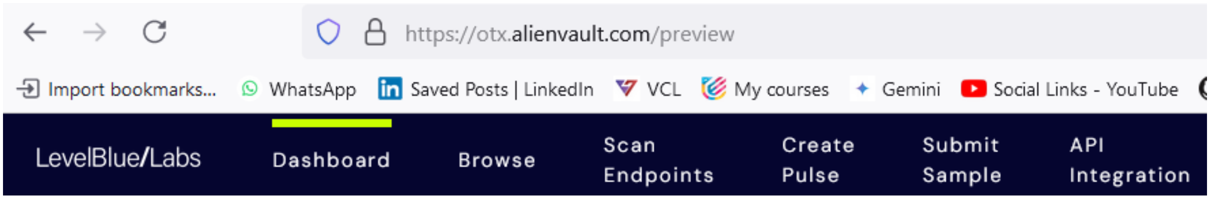


Similarly use both Docguard and VirusTotal to check each sample from Android Malware, Antivirus-Rouges, Browsers, Chrome Extensions, DOS, Downloader, Email worms & Bot Nets, Ransomware, Trojans etc. Remember, these resources may contain malicious content, so proceed with caution and follow best practices for safe browsing.

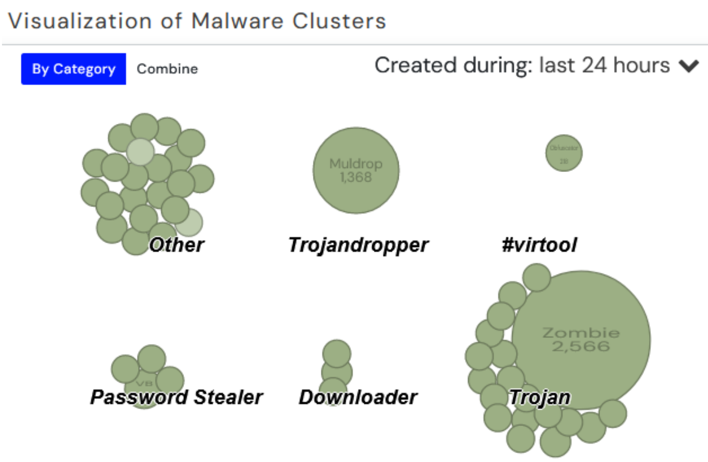
2. AlienVault Open Threat Exchange (OTX)

This is a community-driven threat intelligence platform that allows searching for IOCs like IPs, domains, URLs, and hashes. You can also contribute your own findings to the platform.

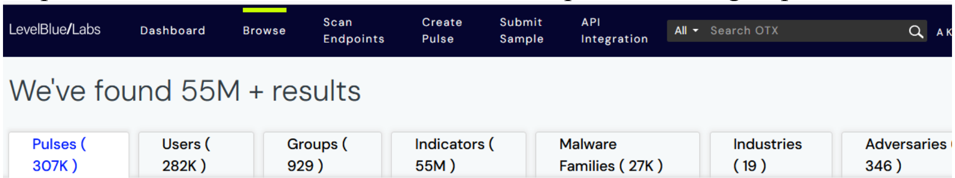
Step 1: Use https://otx.alienvault.com to create a free account, access the dashboard.



Step 2: Check last 24 hours malware families



Step 3: Browse to learn about Malware trends / pulses, users, groups, IoCs, Malware families.

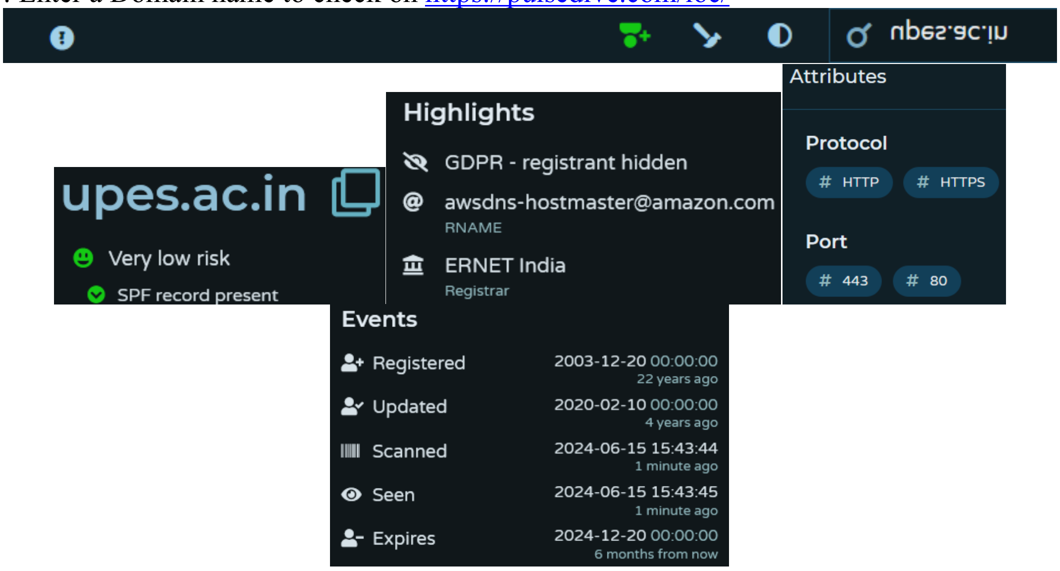


Step 4: Submit a malware sample and view its summary

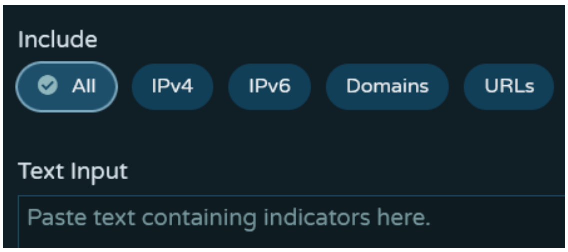
3. PulseDive

Aggregates data from various sources to provide insights on IPs, domains, URLs, and malware samples. It offers free basic features and paid plans for additional functionalities.

Step 1: Enter a Domain name to check on https://pulsedive.com/ioc/



Step 2: Paste text or upload files containing IPs, URLs, and domains to https://pulsedive.com/analyze/. Large, pasted text will be added as a virtual file. Pulsedive will parse, refang, and deduplicate indicators automatically.



4. URLScan

Link: https://urlscan.io

5. **Website Reputation Checker**
Link: https://urlvoid.com

6. IP Address
Link: https://ipvoid.com

7. FileSec
Link: https://filesec.io/#

8. LOLBAS
Link: https://inkd.in/dDa8XgiM

9. GTFOBins
Link: https://inkd.in/dRVzVz87

10. File Hash Check
Link: https://inkd.in/gNqxtn4d

11. Hash Search
Link: **https://inkd.in/eMjdTB2t**

12. Malwares Search
Link: **https://malwares.com**

The best tool for you depends on your specific needs and budget. Here are some factors to consider:

- Features: Consider which features are most important for your work, such as IOC search, threat intelligence feeds, or collaborative analysis capabilities.
- Ease of Use: Some tools are more user-friendly than others. Choose one that is easy to learn and navigate for your team
- Cost: Many free tools offer basic functionalities, while commercial tools provide more advanced features but come with a subscription fee.

Lab activities:

- **Perform the above-mentioned steps and present your work in WORD file with screenshots snipped with relevant details.**
- **Do NOT copy from others, else your submission will be graded ZERO.**